IMPLEMENTATION REPORT

Explore API changes

What changed in the API Playground frontend, Express backend, and PostgreSQL database.

Prepared 17 September 2026 | Workspace: API-PLAYGROUND

> **Result:** The Explore page now places three locally hosted APIs first: Products, Books, and Posts. Their cards link to the backend configured by `VITE_API_URL`. The three existing public API cards remain below them.

1. Request flow

1. `ExplorePage.jsx` requests `GET /api/fake-apis` to build the card grid.
2. The backend reads the `fake_apis` catalog and orders local `/api/...` entries before external entries.
3. Selecting a card requests `GET /api/fake-apis/:id` for its description, sample response, and fetch example.
4. Opening a local card URL calls an Express route that reads one of the seeded PostgreSQL tables.

2. Endpoints added

Resource	List and filter	Single record
Products	`GET /api/products ?category=Home`	`GET /api/products/1`
Books	`GET /api/books ?genre=Science%20Fiction`	`GET /api/books/1`
Posts	`GET /api/posts ?user_id=1`	`GET /api/posts/1`

List endpoints return JSON arrays. A single-record endpoint returns one JSON object, 404 for an unknown ID, and 400 for an invalid ID. Invalid supported filter values return 400. Unexpected database failures return 500.

3. Backend implementation

`backend/src/app.js` mounts the three routes. `sampleDataRoutes.js` defines list and ID handlers for each resource. `sampleDataController.js` validates query values and IDs and maps results to HTTP responses. `sampleDataModel.js` maps each resource to a fixed table and filter column, runs parameterized queries for user input, and sorts list results by ID.

4. Database and example data

`backend/sql/seed_explore_apis.sql` creates three data tables if missing and inserts four records into each. It also adds three rows to `fake_apis`, which supplies the Explore cards and detail panels. Existing catalog entries are preserved.

Table	Fields	Example
`playground_products`	ID, title, category, price, in_stock	Canvas Backpack, Accessories, 49.99, true
`playground_books`	ID, title, author, genre, published_year	The Long Orbit, Mira Sen, Science Fiction, 2021
`playground_posts`	ID, user_id, title, body	Post 1 by user 1: Getting started with APIs

The seed inserts use `ON CONFLICT (id) DO NOTHING` for sample rows and `WHERE NOT EXISTS` for each catalog card, so rerunning the script does not duplicate these records. The catalog entries store relative backend paths such as `/api/products`, categories, tags, descriptions, and representative JSON samples.

The SQL was applied to the PostgreSQL database configured in `backend/.env`. The database had the old catalog rows and no sample tables before seeding. Afterward, the catalog returned six entries: three new local APIs followed by JSONPlaceholder, Fake Store API, and DummyJSON.

5. Explore page changes

- `apiUrl()` expands relative catalog paths with `VITE_API_URL`, which is currently `http://localhost:3000`. Existing absolute external URLs pass through unchanged.
- Card links and the detail panel display full backend URLs for local APIs.
- `usageSnippet()` generates a fetch example with the configured backend URL for local APIs. External cards continue to use their stored snippets.
- The heading changed to **Explore Playground APIs**; the subtitle now describes both local and public APIs.
- `fakeApiModel.js` orders local paths first so new cards are visible at the top, while keeping the earlier public cards.

6. Documentation

`backend/README.md` now explains how to run the seed SQL, lists the new routes and filters, and documents their response behavior.

7. Files changed

File	Purpose
`backend/src/app.js`	Mounts the three new route groups.
`backend/src/routes/sampleDataRoutes.js`	Defines list and ID routes.
`backend/src/controllers/sampleDataController.js`	Validates requests and sends HTTP responses.
`backend/src/models/sampleDataModel.js`	Queries the resource tables in PostgreSQL.
`backend/src/models/fakeApiModel.js`	Places local catalog entries first.
`backend/sql/seed_explore_apis.sql`	Creates and seeds data and catalog rows.
`frontend/src/pages/ExplorePage.jsx`	Displays backend URLs and updated copy.
`backend/README.md`	Documents setup and API usage.

Existing unrelated edit: `backend/src/config/db.js` was already modified before this work and was not changed as part of these API additions.

8. Verification completed

- Frontend production build: `npm.cmd run build` passed.
- Backend modules passed syntax and import checks.
- `GET /api/fake-apis` returned six rows, with Products, Books, and Posts first.
- The three new detail requests returned HTTP 200.
- Products returned four rows; Books filtered to Science Fiction returned two; Posts filtered to user 1 returned two.
- The running backend on port 3000 returned the three local catalog entries first.

9. How to try it

Refresh the Explore page. Open one of the three new cards and select its backend URL. Direct examples:

```
http://localhost:3000/api/products
http://localhost:3000/api/books?genre=Science%20Fiction
http://localhost:3000/api/posts?user_id=1
```

The Explore cards use the configured backend URL. If that setting changes, restart the Vite dev server so the new environment value is loaded.